

Geometric calibration and thermal interpretation in laser melting of IN625: a high-temperature closure diagnosis

Abstract

Geometric calibration gives efficient laser-melting models practical value, yet the thermal meaning of boundary agreement remains incompletely constrained. Solidification interpretation requires distinguishing the location of a melting boundary from the temperature interval and material history behind it. This study examines that tension using a conservative moving-frame enthalpy model of bare-plate IN625 tracks and public AMMT measurements. One effective source factor is fitted to a single length and then fixed for geometric comparisons and a four-corner high-temperature property contrast. The fitted condition remains $19.54\text{ }\mu\text{m}$ wider and $4.88\text{ }\mu\text{m}$ shallower than the reference means. Conductivity holding lengthens the modeled pool by $43.21\text{ }\mu\text{m}$, while sensible-heat-capacity holding shortens it by $6.17\text{ }\mu\text{m}$; their interaction partially offsets the joint response. Most of the conductivity-induced length change is a common rear solidus/liquidus displacement, rather than a comparable enlargement of their separation. Similar rear phase spans at two scan speeds consequently coexist with material passage times of 100.33 and $67.33\text{ }\mu\text{s}$. These comparisons expose the distinct information carried by fusion shape, constitutive response and spatial-to-material-time mapping after one geometric calibration. The resulting diagnosis specifies which thermal interpretations the calibrated model supports and which require an independent thermal observable; it does not identify liquid properties or validate the derived histories.

Keywords: Laser powder bed fusion, IN625, Melt-pool geometry, Enthalpy model, High-temperature properties, Model discrepancy

1. Introduction

Thermal models connect laser processing conditions to the melting and solidification histories that influence material structure in laser powder bed fusion (LPBF). Their usefulness extends beyond estimating the dimensions of a melted track: interpreting a process–structure relationship also requires temperature gradients and the thermal path followed as the material cools. Resolving the coupled laser, phase-change and transport processes over practical build scales remains computationally demanding [1]. Recent finite-element reviews likewise distinguish thermal, mechanical and microstructural modeling across scales, with different material-data and coupling requirements [2]. This creates a persistent tension between affordable heat-transfer calculations, often constrained by accessible melt-pool geometry, and the credibility of the thermal interpretation drawn from them. Thermographic measurements provide additional information on spatial profiles and material histories [3, 4], but geometric agreement remains an important practical constraint. Its meaning depends jointly on the heat-source representation, material response and definition of the measured observable.

The available modeling approaches address different parts of this tension. Powder-resolved thermo-fluid simulations describe recoil pressure, surface-tension-driven flow and evaporation in 316L, while ray-tracing/free-surface calculations have linked multiple reflections to keyhole-induced porosity in Ti6Al4V [5, 6]. Their physical detail is valuable for studying mechanisms that a conduction

model represents only indirectly. Moving-source conduction provides a more economical thermal description. Rosenthal’s solutions established heat-flow and cooling relations for moving sources; later finite-source and nonlinear formulations incorporated temperature-dependent properties and enthalpy storage [7, 8]. Scaling relations based on normalized energy input and dwell-to-diffusion time further reduce the calculation needed to estimate dimensions [9]. Semi-analytical transient conduction has also been used to estimate solidification conditions and relate scan strategies to grain structure in studied AlSi10Mg laser and IN718 electron-beam cases [10]. This hierarchy permits useful simplification, provided the assessment addresses the output and operating regime for which the model is used.

Conduction-based models have achieved substantial geometric agreement under defined conditions. Promoppatum et al. found that both finite-element and Rosenthal approaches reasonably estimated IN718 melt-pool shapes, with differences in absorption sensitivity and high-input behavior [11]. For the AMMT IN625 single-track benchmark, Kollmannsberger et al. combined nonlinear phase-change conduction, measured boundary input and calibrated effective directional conductivity to reproduce the pool dimensions [12]. That result supports the utility of an effective thermal description; it also shows that source and directional material choices can be consequential when a scalar coupling factor alone cannot reproduce the shape. Across materials and regimes, Naderi et al. found that the fidelity of existing depth scaling laws varied and improved with revised, regime-specific relations [13]. Neither successful fitting nor a discrepancy therefore establishes a universal conclusion about conduction models.

Heat-source calibration has extended their practical range. Ross et al. inferred double-ellipsoid parameters from temperatures at the solidification boundary and obtained trends that permit interpolation to intermediate power and speed [14]. Whalen et al. incorporated temperature-dependent and powder-bed properties in an adapted Eagar–Tsai model for 316L, inferred distributions of apparent absorption and porosity, and propagated geometric uncertainty [15]. More recently, Kusano and Watanabe compared four source families for IN738LC and used Bayesian optimization and regression to obtain a conical source applicable within their studied process window [16]. Coleman et al. calibrated a dynamic volumetric source against bare-plate IN625 tracks at 195 W and 800 mm/s over multiple spot sizes, then applied it to an AMB2018-01 layer case [17]. Their formulation includes temperature-dependent, phase-specific properties and latent storage. These achievements make source representation a substantive alternative to material closure when interpreting shape discrepancies.

Additional measurements expose the distinct information carried by geometry, energy coupling and temperature. Myers et al. demonstrated that parameter combinations fitting melt-pool geometry could produce different thermal profiles; two-color measurements helped distinguish those combinations [4]. Calorimetric studies on bare and powder-coated materials show that effective absorption varies with power and surface condition [18]. Simonds et al. compared time-resolved optical coupling with calorimetry during 10 ms stationary 316L welds and found that mass loss affected their energy accounting [19]. Scanning calorimetry and scaling across Ti6Al4V, IN625 and 316L further related coupling and penetration to power, speed and beam size, with marked changes across the conduction–keyhole transition [20]. Those studies provide measurement routes and condition-dependent relationships, rather than an independently known coupling factor for the present tracks. For IN625, Lane et al. distinguished unreflected laser power from substrate heating because plume interactions and mass-energy losses also enter the energy accounting [21]. A fitted source factor consequently carries the meaning of the model and observation used to obtain it. Sensor-to-field mapping also matters: Weeks et al. modeled reflections of thermal emission within concave 316L keyholes to reconcile simulated and camera temperatures [22]. That observation correction addresses a different source of discrepancy from changing the underlying heat-transfer field.

Recent reliability studies make observable-specific assessment explicit. Soares et al. compared an analytical Gaussian conduction model with dual-wavelength imaging and metallography for 316L, IN625 and CoCr single tracks [23]. With fixed properties selected near 80% of the melting temperature and no explicit phase-change treatment, their model gave useful surface comparisons while depth correlation coexisted with substantial scale bias. The result distinguishes correlation across conditions from agreement in a particular dimension. The AMMT measurements likewise define imaging length and metallographic width/depth through different procedures and populations [24]. A geometric constraint is therefore informative only in relation to its observation operator; fitting one dimension does not itself test the remaining shape or a material cooling history.

Material sensitivity and conduction-failure diagnosis are also established research directions. Yang et al. quantified the response of Ti6Al4V thermal fields and dimensions to processing and material parameters using constant-property analytical source models [25]. Hong et al. examined 231 bare-plate tracks with an Eagar–Tsai conduction description, using measured coupling for IN718 and a published absorption relation for IN625 and 316L [26]. Their inversion asks whether measured depth requires physically infeasible absorption, with effective-diffusivity contrasts testing the diagnosis. These studies address parameter sensitivity and model feasibility directly. A complementary need in a nonlinear, geometrically calibrated model is to resolve what a specified closure change does to the local phase-boundary structure, rather than infer that structure from a total dimension or a joint-property response.

This distinction is pertinent when material tables end below the temperatures at which phase change is represented. The IN625 supplier conductivity and sensible-heat-capacity inputs used here terminate below the adopted solidus [27]. Continuing those tables introduces assumptions affecting both heat redistribution and storage. In an enthalpy formulation, sensible heat capacity and latent-inclusive storage are different quantities; changing both continuations together can conceal their separate responses. Similarly, movement of the rear solidus and liquidus positions together changes pool length without requiring comparable growth of the interval between them. That spatial phase interval and the time spent traversing it are also distinct, because material passage through a quasi-steady translating field depends on scan speed. Separating these responses makes the thermal meaning of a geometric sensitivity more explicit.

Here, a nonlinear moving-frame enthalpy model examines these distinctions for three AMMT bare-plate IN625 conditions. One effective source factor is fitted to the B surface-length proxy. Width and depth are excluded from that fit, while A and C provide retrospective, nonblind comparisons with the factor fixed. At B, a four-corner contrast separately replaces linear continuation by last-value holding for conductivity and sensible heat capacity, retaining the source factor and phase law without recalibrating the branches. The analysis separates their finite effects and interaction, decomposes rear solidus/liquidus displacement from phase-span change, and maps that span to passage time and interval-mean cooling rate using scan speed. Numerical verification supports these model comparisons; the derived histories remain distinct from independent thermal validation. The study thus provides a controlled diagnosis of thermal interpretation after one geometric calibration, with explicit distinctions between fusion-shape discrepancy, closure response and material time.

2. Materials and methods

2.1. Study design, experimental referent and data roles

The study tests how one-target geometric agreement relates to fusion shape, constitutive sensitivity and derived material history. First, B length calibrates one effective source factor. Width and depth, which are excluded from that objective, then assess geometric discrepancy; A and C assess the fixed-parameter condition response. Next, a four-corner contrast at B separates conductivity and

sensible-storage continuation while retaining power, speed, beam, density, phase thresholds, latent heat, mesh and observation operators. Decomposing the rear solidus and liquidus positions distinguishes translation of the thermal tail from widening of the phase interval. Finally, mapping the rear spatial interval through scan speed distinguishes geometric extent from material passage time.

The referent is constant-speed laser scanning on bare IN625 plate under three public AMMT conditions (Table 1). These are single tracks without a powder layer. The three conditions cover increasing speed; A also has a different power, whereas B and C isolate a speed change at fixed nominal power. The reported beam diameter is $D_{4\sigma} = 170 \mu\text{m}$. The circular Gaussian reconstruction below uses a radius parameter $w = 85 \mu\text{m}$; it does not reproduce the measured two-dimensional beam map.

Table 1: Operating conditions and fixed-source energy scales. The effective factor $\eta = 0.2890548519$ is fitted to B length only. $E_\ell = \eta P/v$ and w/v are model input scales.

Condition	P (W)	v (m/s)	E_ℓ (J/m)	w/v (μs)
A	137.9	0.4	99.65	212.50
B	179.2	0.8	64.75	106.25
C	179.2	1.2	43.17	70.83

The length reference is the AMMT condition mean reported by Lane et al. [24]. Current width and depth means are taken from NIST challenge Table 2 for the 100 μs -track microscopy classes [28]. The latter differ in aggregation and display precision from the subsequently published cross-section means in Lane et al., which combine 100 and 20 μs tracks. Length and cross-section references are therefore condition-level quantities from separate observation populations, rather than a joint three-dimensional measurement of one track.

The B length is the sole scalar calibration target. B width and depth are diagnostics after calibration, and A/C are retrospective, nonblind comparisons with η frozen. All public targets were available during development. Expanded uncertainties U with coverage factor $k = 2$ are taken separately from Lane’s uncertainty budgets. For width and depth these are descriptive measurement scales from the published combined-class analysis, not uncertainty values jointly estimated with the current decimal NIST means. The ratio δ/U , where δ is the signed model-reference difference, is used only to report discrepancy scale. It is not a significance test or a combined numerical, input, calibration and measurement uncertainty statistic.

2.2. Conservative enthalpy model in the laser frame

The laboratory-frame conduction balance is

$$\frac{\partial H}{\partial t} = \nabla \cdot [k(T)\nabla T], \quad H(T) = \rho \left[\int_{T_0}^T c_p(s) ds + \mathcal{L}f(T) \right]. \quad (1)$$

The adopted constants are $\rho = 8440 \text{ kg m}^{-3}$, $\mathcal{L} = 280000 \text{ J kg}^{-1}$, $T_0 = 25 \text{ }^\circ\text{C}$, $T_s = 1290 \text{ }^\circ\text{C}$ and $T_l = 1350 \text{ }^\circ\text{C}$. The equilibrium phase fraction is

$$f(T) = \begin{cases} 0, & T < T_s, \\ (T - T_s)/(T_l - T_s), & T_s \leq T < T_l, \\ 1, & T \geq T_l. \end{cases} \quad (2)$$

Specific heat and conductivity are linearly interpolated between the actual input knots in Table A.4 in the reproducibility appendix. The conductivity table contains 15 points and ends at 982 $^\circ\text{C}$; the

12-point specific-heat table ends at 1093 °C. These are the endpoints loaded by the executed material routine. The supplier labels its typical specific heats as calculated and conductivity as measured [27]; these are general alloy data, not measurements on the benchmark coupon. Density and the melting interval use that bulletin, while the adopted latent heat follows the benchmark modeling setup [12]. The sensible enthalpy integral is evaluated piecewise analytically; adding the phase contribution preserves a monotone temperature–enthalpy relation and permits analytic inversion. The latent contribution to effective specific heat inside the phase interval is 4666.67 J kg⁻¹ K⁻¹.

Figure 1 separates tabulated interpolation from the high-temperature assumptions: both endpoints lie below the adopted solidus.

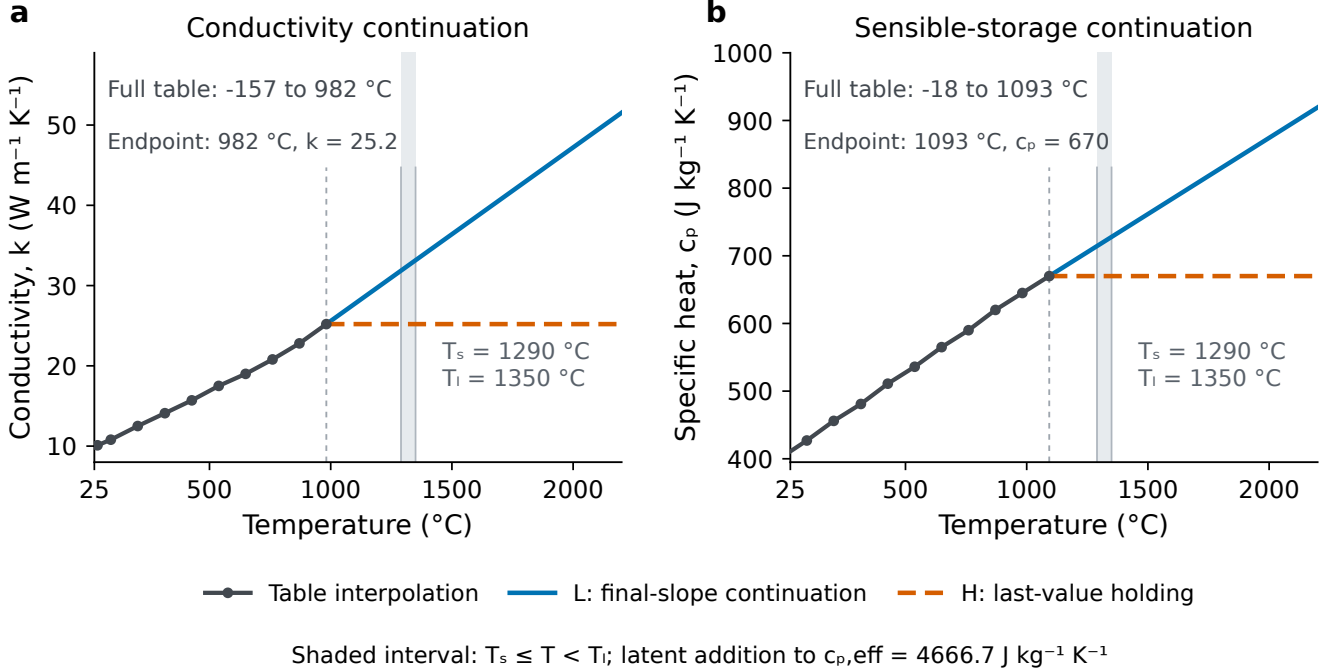


Figure 1: Actual IN625 property inputs and the two continuation choices. (a) Conductivity and (b) sensible specific heat use the knots loaded by the executed material routine, with common piecewise-linear interpolation below the respective endpoints. The final knots are 982 °C and 25.2 W m⁻¹ K⁻¹ for k , and 1093 °C and 670 J kg⁻¹ K⁻¹ for c_p . L continues the final tabulated secant; H holds the final value. The displayed window is 25–2200 °C; all 15 conductivity and 12 specific-heat source rows, including those below 25 °C, are retained in the accompanying data and used in interpolation. Shading marks the adopted 1290–1350 °C phase interval. The latent addition to effective specific heat within this interval is 4666.7 J kg⁻¹ K⁻¹ and is stated separately from sensible c_p . The supplier describes conductivity as measured and specific heat as calculated typical alloy data [27]; the plotted high-temperature continuations are modeling choices.

For a locally quasi-steady translating field, define $\xi = x - vt$ and $\mathbf{u} = (-v, 0, 0)$. Equation (1) becomes

$$\nabla \cdot (\mathbf{u}H - k\nabla T) = 0. \quad (3)$$

The term $\mathbf{u}H$ represents stationary material passing through the laser coordinate frame. No liquid momentum equation or melt convection is included. The assumptions also exclude evaporation and free-surface deformation. Consequently, coordinate transport is not evidence of liquid flow.

The Gaussian surface input at $z = 0$ is

$$q(\xi, y) = \frac{2\eta P}{\pi w^2} \exp\left[-\frac{2(\xi^2 + y^2)}{w^2}\right]. \quad (4)$$

With outward normal $\mathbf{n}$, the top condition is $-k\nabla T \cdot \mathbf{n} = -q$; the Gaussian supplies inward heat. Rectangle-wise error-function integration supplies each top-face power. Its full-plane integral is ηP and the transverse half-domain receives $\eta P/2$. The finite-volume implementation deposits the integrated boundary input into top control volumes. At positive ξ , incoming material has $H = 0$; at negative ξ , diffusive gradient vanishes and enthalpy leaves advectively. The $y = 0$ plane is symmetric. Outer transverse and bottom boundaries are adiabatic. Radiation and ambient convection are omitted from the solved balance.

2.3. Finite-volume implementation and nonlinear solution

The domain is $\xi \in [-1, 0.35]$ mm, $y \in [0, 0.36]$ mm and $z \in [-0.30, 0]$ mm. A graded orthogonal mesh contains 902664 cells. Near-source cell widths are $4 \times 3 \times 1 \mu\text{m}$ within $\xi \in [-0.48, 0.16]$ mm, $y \in [0, 0.12]$ mm and $z \in [-0.06, 0]$ mm. Outer widths increase by a factor of 1.22 up to $25 \mu\text{m}$. The half-domain mesh has $199 \times 56 \times 81$ cells. The solved problem is steady; retained warm fields provide nonlinear initial guesses rather than a transient thermal initial condition.

FiPy 4.0.3 [29] assembles diffusion and exponential coordinate-convection terms. The scaled unknown is $U = H/C_0$, with $C_0 = \rho(402 \text{ J kg}^{-1} \text{ K}^{-1}) = 3.39288 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$. On an internal face,

$$D_f = \frac{k_f}{C_0} \frac{\Delta T}{\Delta U}, \quad (5)$$

where k_f is the harmonic face conductivity and D_f has units of $\text{m}^2 \text{s}^{-1}$. For $|\Delta U| \leq 10^{-8} \text{ K}$, the implementation uses the positive derivative limit $\Delta T/\Delta U = C_0/[\rho(c_p + \mathcal{L}f')]$ at the first adjacent cell. This yields the corresponding conductive flux while conserving enthalpy; U is a scaled enthalpy, not temperature. SciPy Anderson acceleration [30] solves the nonlinear fixed-point residual; the inner linear systems use GMRES with tolerance 2×10^{-9} and restart 80, preconditioned by PyAMG Ruge–Stuben classical multigrid. Detailed acceleration and linear iteration controls are provided in the reproducibility appendix. Final acceptance uses the assembled nonlinear balance and the full-solve correction described below. The executed environment records Python 3.13.5, NumPy 2.5.3, SciPy 1.18.1 and PyAMG 5.3.0, with one BLAS thread.

The diffusion and exponential-convection objects are recreated after each coefficient update. This step removes dependence on cached initial stencil weights found during implementation checks. Nonlinear convergence requires an assembled absolute cell-residual sum normalized by absorbed half-power below 2×10^{-5} , signed global balance magnitude below the same scale, and a further frozen-coefficient solve changing temperature by less than $0.002 \text{ }^\circ\text{C}$. Local residual and global balance are retained separately because opposing local errors can cancel in a global sum.

2.4. Geometry and passage-history operators

Model length L is the distance between two solidus crossings of the recovered surface symmetry-centerline temperature. Crossings are interpolated, must enclose the source, and must form one interval without domain truncation. Surface recovery satisfies

$$K(T_{\text{surf}}) - K(T_{\text{cell}}) = q \Delta z/2, \quad K'(T) = k(T), \quad (6)$$

and the symmetry-plane value uses the even quadratic extension $T(0) = T(y_1) + [T(y_1) - T(y_2)]y_1^2/(y_2^2 - y_1^2)$ from the first two positive- y cell centers. The recovered surface temperatures and symmetry extension are applied before the threshold crossings are extracted.

Fusion-zone width and depth are extracted from the source-connected component of

$$T_{\text{max}}(y, z) = \max_{\xi} T(\xi, y, z) \geq T_s. \quad (7)$$

Each ξ slice is first reconstructed at the surface and symmetry plane, then bilinearly interpolated onto observation coordinates, and only then maximized over passage. Taking a nodal maximum before interpolation creates a different envelope. Width is twice the positive- y extent and depth is the negative of the lowest z . The $0.5 \mu\text{m}$ observation pitch samples interpolated geometry; it does not represent the resolution of the heat-field mesh. The geometric figure labels L_s , W_f and D_f correspond to the text's L , W and D , respectively: L_s is the surface length proxy, while W_f and D_f are fusion-envelope extents.

Lane's imaging length uses a radiance-profile second-derivative minimum for the rear solidification location and a calibrated front threshold [24]. The present two-solidus-crossing proxy does not replay radiance, emittance, pixel integration, viewing geometry or motion blur. Its fitted η is therefore an effective source/model/operator parameter. Width and depth refer instead to a maximum-temperature fusion envelope, compared with metallography.

Let ξ_s^- and ξ_l^- denote the rear centerline solidus and liquidus crossings. Define the rear spatial phase span, passage time and interval-mean cooling rate as

$$\ell_m = \xi_l^- - \xi_s^-, \quad \tau_m = \frac{\ell_m}{v}, \quad \bar{T} = \frac{T_l - T_s}{\tau_m}. \quad (8)$$

These are derived from one steady temperature solution. The passage time is a coordinate transformation, not a second independently predicted observable. The cooling interval is 1350–1290 °C and differs from the below-solidus example interval used by Lane.

2.5. Calibration and high-temperature continuation contrast

The baseline continues the final tabulated slope of each property above its endpoint. A safeguarded secant solves $L_B(\eta) - 359 \mu\text{m} = 0$ in the initial bracket $[0.2, 0.4]$, with proposals restricted to the inner 90% of the current bracket and an accepted absolute length residual below $0.5 \mu\text{m}$. It yields $\eta = 0.2890548519203547$, with a residual of $+0.158 \mu\text{m}$. Calibration trials and the local sensitivity conversion are reported in the appendix. A and C retain the final value, the same beam and phase law, and the same continuation rules.

At fixed B, two binary factors replace linear continuation (L) by holding the last tabulated value (H), independently for k and c_p . H holds $k = 25.2 \text{ W m}^{-1} \text{ K}^{-1}$ above 982 °C and $c_p = 670 \text{ J kg}^{-1} \text{ K}^{-1}$ above 1093 °C. The final table secants used for L are $(25.2 - 22.8)/(982 - 871) = 0.0216216 \text{ W m}^{-1} \text{ K}^{-2}$ and $(670 - 645)/(1093 - 982) = 0.225225 \text{ J kg}^{-1} \text{ K}^{-2}$. The changes apply above each table endpoint, including the solid range between the endpoint and T_s . Each conductivity branch changes k , its primitive K and the surface inverse together; each storage branch changes c_p , H and $T(H)$ together. The first letter labels conductivity. The LL solution is the B baseline; HH is a retained joint-closure comparison; HL and LH supply the two mixed corners. No corner is recalibrated. Across A/B/C and these four corners, sharing B=LL gives six unique production runs.

For any output Q , define the baseline-anchored finite effects

$$\begin{aligned} \delta_k &= Q_{\text{HL}} - Q_{\text{LL}}, & \delta_{c_p} &= Q_{\text{LH}} - Q_{\text{LL}}, & (9) \\ I_{k,c_p} &= Q_{\text{HH}} - Q_{\text{HL}} - Q_{\text{LH}} + Q_{\text{LL}}, & \Delta_{\text{joint}} &= Q_{\text{HH}} - Q_{\text{LL}} = \delta_k + \delta_{c_p} + I_{k,c_p}. & (10) \end{aligned}$$

The conditional conductivity effect at held c_p is $Q_{\text{HH}} - Q_{\text{LH}} = \delta_k + I_{k,c_p}$; the conditional storage effect at held k is $Q_{\text{HH}} - Q_{\text{HL}} = \delta_{c_p} + I_{k,c_p}$. Reporting both conditions checks whether a direction changes with the other factor. These deterministic contrasts distinguish two continuation choices within the same model. They are not estimates from replicated material experiments and have no sampling-significance interpretation.

The retained face transport diagnostic is

$$|\text{Pe}_f| = \frac{|\mathbf{u} \cdot \mathbf{n}_f| d_f}{D_f}, \quad (11)$$

where d_f is the adjacent cell-center separation. Unweighted medians are taken over internal axial faces adjacent to at least one cell with $T_s \leq T < T_l$. The diagnostic compares coordinate transport with diffusion in the discretized field; its selected face population can change between solved branches.

The discrete rear hot-region budget selects cells with $T \geq T_s$ and $\xi < 0$. Net outward diffusive power is the sum of signed conductive face powers on the boundary of this selected cell set; internal face contributions cancel. The region is selected separately in each field, so its boundary and sampled faces change with the continuation. This accounting diagnostic is interpreted alongside the matched-coordinate centerline thresholds rather than as a fixed-region causal contrast.

2.6. Numerical evidence for the reported comparisons

Verification and physical comparison are treated separately [31]. An analytical constant-property moving Gaussian reference gives $L/W/D = 407.535/153.293/34.836 \mu\text{m}$; the corresponding finite-volume values differ by +0.139%, +0.190% and -0.499% . This checks the source, frame and extraction for a simpler problem. Its separate controls and reference integral are given in the appendix; it does not establish nonlinear phase-change accuracy.

At frozen calibrated η , axial refinement from 4 to 2 μm changes B length, width and depth by +0.361, -0.009 and +0.041 μm , respectively. The corresponding C changes are +0.199, -0.029 and +0.045 μm . A/B domain enlargement changes these quantities by less than $3 \times 10^{-6} \mu\text{m}$. These are measured sensitivity contrasts rather than certified error bounds. There is no branch-specific three-dimensional asymptotic refinement sequence. In particular, submicrometre interaction effects are reported descriptively. An older transient comparison at a different B calibration stage supplies only approximate local steady consistency. Fixed-field, blackbody top-loss estimates for the B corners are 0.099–0.180% of absorbed power; they screen the loss scale without solving feedback into the field.

All six retained production fields were inspected for finite temperature, bounded phase fraction, qualified nontruncated geometry, integrated source power and executed-source/dependency consistency. Their normalized absolute residuals range from 6.90×10^{-8} to 1.31×10^{-7} ; signed global balance magnitudes are below 6.1×10^{-11} , and full-solve temperature corrections are below $2.7 \times 10^{-5} \text{ }^\circ\text{C}$. Thus the final results satisfy the recorded nonlinear gates.

The original numerical case retains executed solver snapshots, input manifests, temperature arrays, calibration history and primary-observer code for the six solutions. Source-power integration, threshold crossings, primary passage geometry and finite-contrast algebra were independently reproduced from these artifacts.

3. Results

3.1. Length fitting leaves condition-dependent shape discrepancy

The one-target calibration reproduces the B length proxy while leaving a systematic width/depth residual at B and C (Table 2 and Fig. 2). The fitted length is 359.16 μm , but the B width exceeds the nominal reference by 19.54 μm and depth falls below it by 4.88 μm . Their separate expanded-uncertainty-scale ratios are +3.10 and -1.23 . Thus length fitting does not reproduce the remaining B shape quantities.

A differences are smaller than their respective reported uncertainty scales. In C, the model is 49.37 μm shorter, 23.05 μm wider and 7.56 μm shallower than the nominal references. The wide/shallow pattern is also apparent in the predicted aspect ratios $W/D = 3.79, 4.60, 5.85$ for

Table 2: Condition-level geometry comparisons in μm . U is the separately sourced Lane expanded uncertainty ($k = 2$). Reference width/depth means are current NIST 100 μs -track classes. B length is fitted. δ/U uses only the experimental scale, without combined model uncertainty.

Condition	Quantity	Model	Reference	U	δ/U
A	L	309.89	300	11.91	+0.83
	W	153.36	147.9	6.42	+0.85
	D	40.47	42.5	4.49	-0.45
B	L (fitted)	359.16	359	21.26	+0.01
	W	143.04	123.5	6.30	+3.10
	D	31.12	36	3.97	-1.23
C	L	320.63	370	32.57	-1.52
	W	129.05	106	4.48	+5.14
	D	22.04	29.6	3.26	-2.32

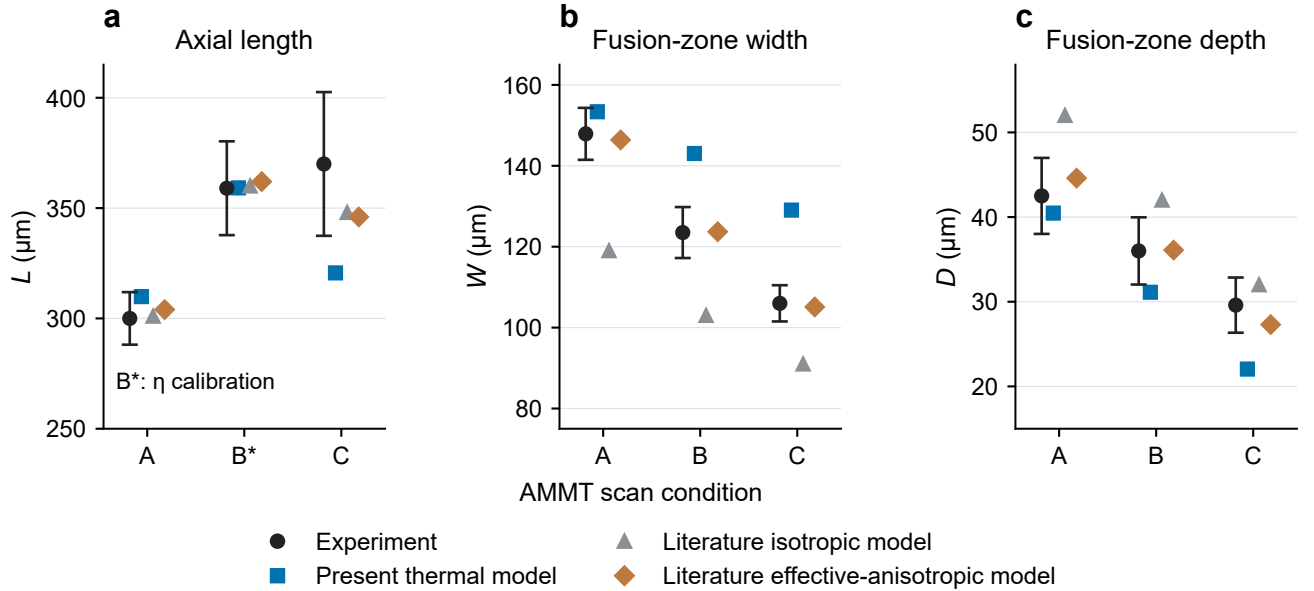


Figure 2: Geometry comparison with condition-level reference means and the published isotropic and effective-anisotropic calculations of Kollmannsberger et al. [12]. Experimental bars show separately sourced Lane $U(k = 2)$ scales; width/depth centers use current NIST means. B length is calibrated in the present model. Published-model points provide context: beam representation, phase law, parameter calibration and observation definitions differ, so this is not a controlled ranking of solvers.

A/B/C, compared with condition-level reference ratios 3.48, 3.43, 3.58. These ratios combine separate class means; they are not specimen-wise aspect-ratio measurements.

At fixed power from B to C, a 50% speed increase reduces absorbed line energy by one third. Model width decreases by 9.78% and depth by 29.16%; the nominal reference changes are 14.17% and 17.78%. The modeled W/D increases by 27.36% from B to C. The shape discrepancy therefore concerns the relative contraction of transverse and depth extents as well as their absolute values. A–B changes cannot be attributed to speed alone because power also changes.

Nominal experimental length increases by 11 μm from B to C, whereas model length decreases by 38.53 μm . The 11 μm increase is smaller than the 38.89 μm zero-correlation quadrature of the two reported expanded-uncertainty scales. The data therefore do not establish a statistically resolved

opposite experimental length trend. C's individual discrepancy remains distinct from that trend comparison. Retained axial changes below $0.37\text{ }\mu\text{m}$ are far smaller than the main B/C residuals; refining that direction alone is unlikely to remove the observed shape pattern.

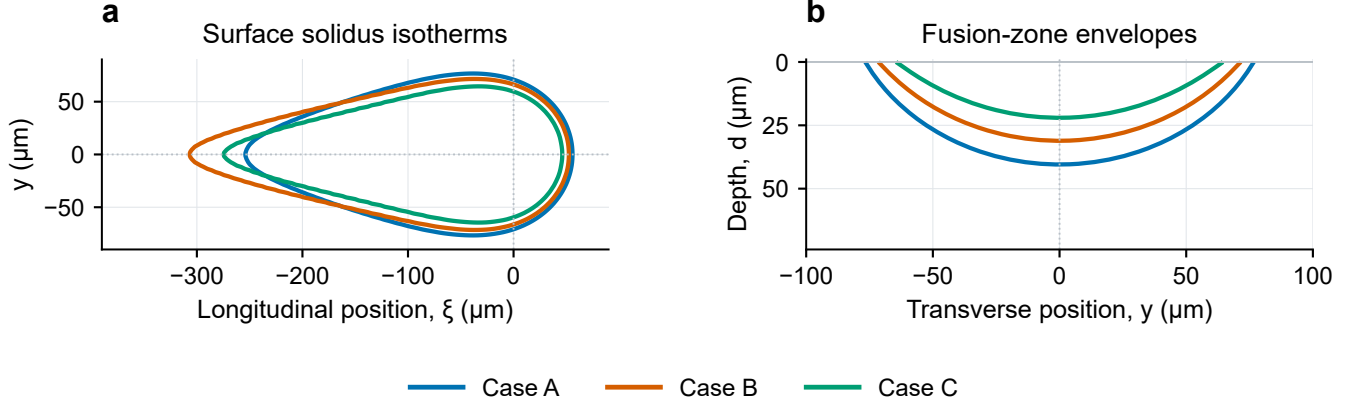


Figure 3: Model geometry for A, B and C. (a) Surface solidus contours in laser-frame coordinates. (b) Reconstructed maximum-temperature fusion envelopes in transverse/depth coordinates. Fusion envelopes use slice reconstruction and interpolation before passage maximization. The fields are model outputs, not microscopy or inferred liquid-flow maps. Both panels use the fixed calibrated source factor.

The common physical axes make the decrease in penetration relative to transverse width apparent. The surface contour and fusion envelope answer different geometric questions: the first determines the surface length proxy, while the second follows the maximum-temperature material passage through the field.

3.2. Conductivity and storage continuations produce opposing tail responses

The four-corner results are shown in Table 3 and Fig. 4. Holding conductivity at its last table value increases length from 359.16 to $402.37\text{ }\mu\text{m}$ at linearly continued c_p . Holding c_p instead decreases length to $352.99\text{ }\mu\text{m}$. Holding both yields $393.44\text{ }\mu\text{m}$. The joint increase is therefore

$$\Delta L_{\text{joint}} = 43.21 - 6.17 - 2.76 = 34.28\text{ }\mu\text{m}. \quad (12)$$

The heat-capacity response and interaction offset part of the conductivity response. A comparison of LL and HH alone would miss this decomposition. The signs persist at the other factor level: conductivity holding increases length by $40.45\text{ }\mu\text{m}$ when c_p is held, and storage holding decreases length by $8.93\text{ }\mu\text{m}$ when k is held. Conductivity therefore extends the tail and storage holding shortens it in both tested settings, with the interaction changing the magnitudes.

Conductivity holding moves the rear solidus and liquidus by -42.49 and $-43.50\text{ }\mu\text{m}$, while the front solidus moves by $+0.73\text{ }\mu\text{m}$. The rear solidus displacement accounts for 98.3% of the conductivity-induced length increase; the front contributes the remaining 1.7%. The rear phase span changes by only $-1.01\text{ }\mu\text{m}$. Heat-capacity holding moves the rear positions forward by $+6.22$ and $+6.53\text{ }\mu\text{m}$, changing their separation by $+0.31\text{ }\mu\text{m}$. The large length response thus predominantly translates both rear phase boundaries together. It does not represent comparable broadening of the mushy interval.

At $1320\text{ }^\circ\text{C}$, conductivity changes from 32.51 to $25.20\text{ W m}^{-1}\text{ K}^{-1}$ under holding, a reduction of 22.48%. Sensible specific heat decreases from 721.13 to $670\text{ J kg}^{-1}\text{ K}^{-1}$, or 7.09%. With the latent contribution included, effective specific heat decreases from 5387.79 to $5336.67\text{ J kg}^{-1}\text{ K}^{-1}$, or 0.95%,

Table 3: Fixed-B deterministic continuation contrast. The first letter labels k , the second c_p ; L means linear continuation and H last-value holding above the respective endpoints. No branch is recalibrated. Rounded geometry and effects are in μm . Small transverse effects lack branch-specific refinement evidence.

Corner or contrast	L	W	D
LL	359.16	143.04	31.12
HL	402.37	143.15	30.09
LH	352.99	143.72	31.61
HH	393.44	143.79	30.80
δ_k	+43.21	+0.11	-1.03
δ_{c_p}	-6.17	+0.68	+0.49
I_{k,c_p}	-2.76	-0.04	+0.21
Δ_{joint}	+34.28	+0.75	-0.32

and enthalpy from the cold reference decreases by 0.66%. This matched-temperature comparison describes the adopted mushy interval. A retained local face Péclet diagnostic for the mushy ξ faces increases in median from 4.44 in LL to 5.77 in HL, consistent with reduced diffusion relative to coordinate enthalpy transport. The diagnostic is consistent with a conductivity-dependent shift in the model’s transport balance. The corresponding retained rear hot-region outward diffusive flux rises from 14.724 to 14.913 W. Since that region is selected by temperature and changes with the solved state, conductivity holding cannot be equated with a measured reduction in its integrated outward heat flow. The branch also changes surface recovery consistently with k .

The current-mesh depth contrast illustrates cancellation. Conductivity holding changes it by $-1.03 \mu\text{m}$; heat-capacity holding contributes $+0.49 \mu\text{m}$ and the interaction $+0.21 \mu\text{m}$, leaving a net $-0.32 \mu\text{m}$. All corners remain wider and shallower than the B reference. The conductivity depth effects remain negative at both storage levels (-1.03 and $-0.82 \mu\text{m}$), whereas the storage depth effects remain positive at both conductivity levels ($+0.49$ and $+0.71 \mu\text{m}$). These two continuation options therefore do not repair the baseline shape discrepancy. The main 34–43 μm length response substantially exceeds the retained baseline axial sensitivity; the 0.04 μm width interaction has no corresponding precision demonstration and is not interpreted as a robust fine ranking.

Across the four corners, the rear phase span lies between 79.25 and 81.17 μm while total length ranges from 352.99 to 402.37 μm . The contrast consequently changes the location of the rear phase interval much more than its spatial separation. This is the main comparison supplied by the retained centerline profiles; the small span ordering remains a current-mesh result.

3.3. Spatial phase span and material cooling time separate across speed

On the present mesh, the rear centerline phase spans are 48.30, 80.26 and 80.79 μm for A/B/C (Fig. 5). The B/C spans differ by only 0.53 μm , yet their transformed passage times are 100.33 and 67.33 μs . The C span is 0.66% larger, while the passage time is 32.90% shorter because speed increases from 0.8 to 1.2 m/s. The corresponding interval-mean cooling rate rises from 0.598 to 0.891 MK s^{-1} ; A gives 120.74 μs and 0.497 MK s^{-1} . Thus the B-to-C interval-mean cooling rate increases by 49.02% despite only a 0.53 μm difference in the spatial phase span.

The result is a distinction between spatial structure and exposure time, rather than an independent cooling-rate validation. It follows directly from the coordinate relation in Eq. (8). A phase-span image alone cannot specify the material thermal interval without the scan speed. The inferred histories refer to the centerline and adopted equilibrium phase law; they do not predict grain morphology, segregation or final microstructure.

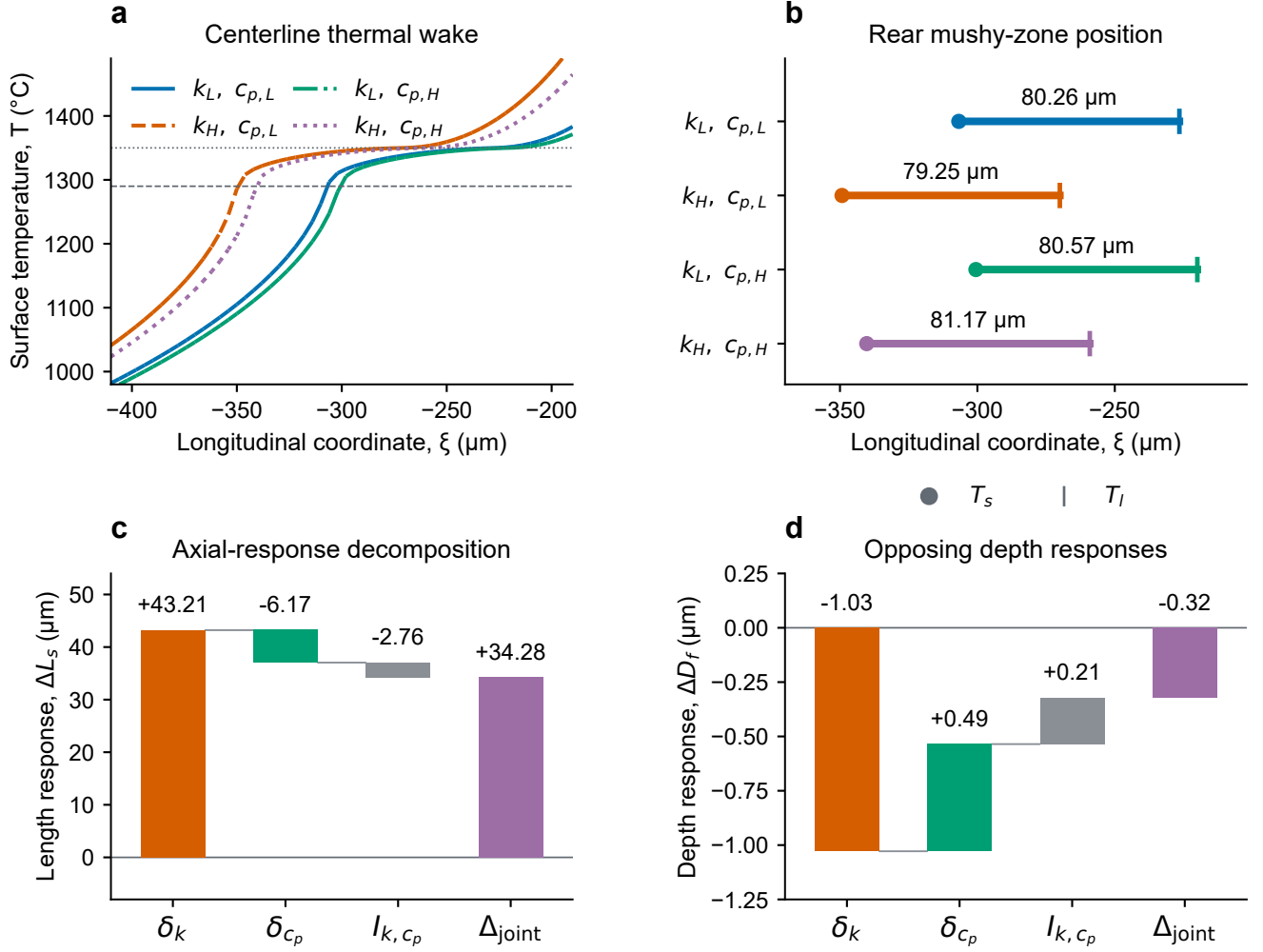


Figure 4: Four-corner continuation comparison at fixed B: rear centerline temperature profiles, phase-boundary positions and spatial span, and baseline-anchored geometry effects. The same phase thresholds, source factor and mesh are retained. Profiles are plotted from original field values with segment clipping to the displayed window. Factorial effects are deterministic finite contrasts, not statistical effect estimates. Large rear translation dominates the length response; joint depth change conceals opposing single-factor contributions.

The operating-condition comparison therefore separates a change in the modeled fusion shape from a change in material passage time. The latter follows from the rear thresholds and speed; its experimental counterpart is not included in the geometric comparisons.

4. Discussion

4.1. Geometric agreement as a conditional constraint

A length calibration constrains the combination of source strength, constitutive assumptions and observation operator that reproduces one selected boundary extent. In the present model, agreement with B length leaves a wider and shallower fusion envelope than the reference, and the relative contraction of width and depth differs from the nominal B-to-C response (Fig. 2). The residual concerns the relative transverse and depth response and is compatible with differences in energy deposition,

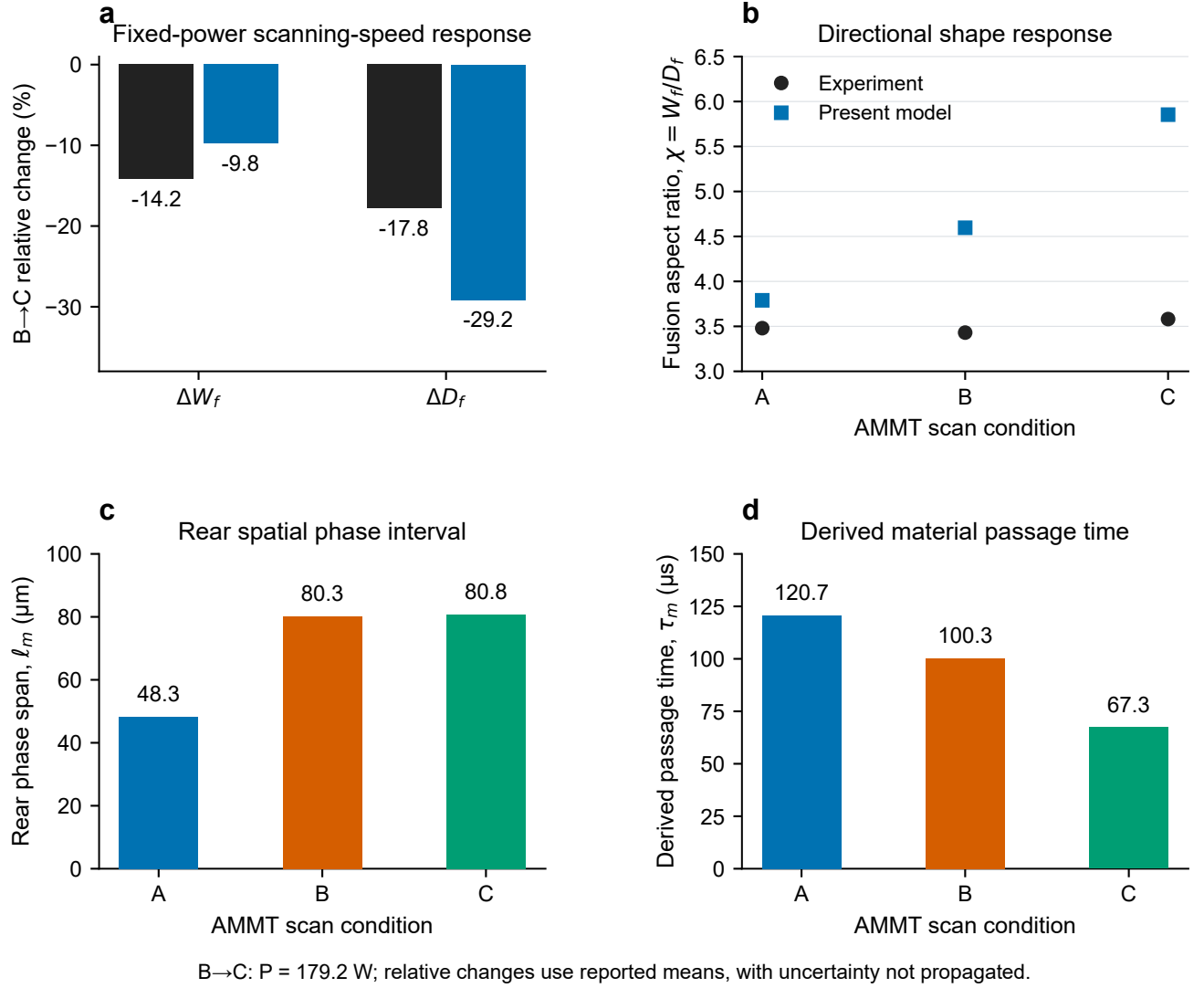


Figure 5: Operating-condition response. (a) Nominal B-to-C percentage changes in width and depth at fixed power, comparing model and condition-level reference means; uncertainty is not propagated. (b) Model and reference class-mean aspect ratios. (c) Model rear liquidus-to-solidus spatial spans. (d) Derived material passage times. The model histories share one steady solution and should not be counted as independent validation observables. Interval-mean cooling rates discussed in the text are derived over 1350–1290 °C; they are not plotted here.

transport closure or observation mapping. Fitting the source factor to length does not supply an independent constraint on penetration, rear phase separation or the underlying temperature profile.

This distinction has a direct precedent in Myers et al. [4]. Their bare-plate 316L FLOW-3D calculations fitted Fresnel and accommodation coefficients simultaneously to cross-sectional area and width-to-depth ratio across six power–speed combinations. Parameter combinations with similar geometric agreement produced different surface temperatures; two-color profiles provided an additional discriminating observable. That model included melt motion and vaporization but used constant surface tension. Its finding establishes a geometry–temperature ambiguity under those conditions, rather than attributing the present residual to the same parameters. Here only the baseline length is fitted, and the continuation branches retain its source factor. Their differences consequently test response after one calibration, rather than demonstrate different thermal histories among equally refitted models.

Kollmannsberger et al. [12] provide the closer AMMT geometry comparison. With the measured laser input constraining the source shape, scalar absorption calibration left appreciable width/depth errors; effective directional conductivity calibrated at B improved the A/C geometry comparisons. Their temperature-dependent, latent-inclusive formulation already addresses nonlinear material behavior. The contrast with the present isotropic Gaussian calculation shows that source representation and directional closure can change geometric agreement. It does not rank the solvers, because the beam, material continuation, phase treatment and extraction differ. Importantly, their study explicitly limited the validity of internal melt-pool temperatures despite improved geometry.

Recent reliability studies reinforce the need to distinguish observables. Soares et al. [23] compared a Gaussian conduction model with thermal-camera and metallographic measurements for powder tracks in 316L, IN625 and CoCr. Properties were fixed near $0.8T_{\text{melt}}$, with powder corrections and no explicit phase change. Good correlation coexisted with biased surface area and regime-dependent depth discrepancy. Their camera calibration used melting and metallographic information, so these channels were not wholly independent. The present bare-plate enthalpy model asks a different question, but shares the lesson that agreement in one dimension or correlation across conditions does not establish agreement of the full field.

4.2. The information gained by fixing the calibrated source

Keeping the source fixed makes the closure contrast interpretable: each change is a response to a specified continuation choice, without an absorption adjustment absorbing part of it. The joint length increase of $34.28 \mu\text{m}$ combines a conductivity contribution of $+43.21 \mu\text{m}$, a sensible-storage contribution of $-6.17 \mu\text{m}$ and an interaction of $-2.76 \mu\text{m}$. Conductivity holding lengthens the pool at both storage settings, whereas storage holding shortens it at both conductivity settings. A joint property replacement would conceal these opposing responses. The small joint depth change similarly conceals cancellation, and none of the four corners repairs the B wide/shallow pattern (Table 3).

The rear profiles explain what the length response represents. Both rear phase boundaries move together by approximately $43 \mu\text{m}$ under conductivity holding, while their separation changes by approximately $1 \mu\text{m}$ (Fig. 4). Rear solidus displacement supplies 98.3% of the length change. Thus the dominant response is a translation of the thermal tail relative to the source, rather than a comparable enlargement of the axial phase interval. Tracking solidus and liquidus separately exposes information that total length alone suppresses.

Material sensitivity itself is established. Yang et al. [25] varied constant properties and source parameters in analytical Ti6Al4V models and assessed their contributions to geometry and temperature. Whalen et al. [15] incorporated temperature-dependent properties and powder conditions into an adapted Eagar–Tsai model, inferred distributions of apparent absorption and porosity, and propagated geometry uncertainty for 316L. The present deterministic four-corner comparison instead separates two missing-range functions in one nonlinear enthalpy model and connects their finite interaction to rear-threshold motion. Its effect magnitudes are conditional responses, rather than experimental property estimates or probabilistic uncertainty bounds.

4.3. Transport and storage: a conditional mechanism

Latent-inclusive storage helps explain the modest sensitivity to sensible heat capacity in the adopted phase interval. The fixed linear phase law adds $4666.67 \text{ J kg}^{-1} \text{ K}^{-1}$ to effective specific heat between solidus and liquidus. At 1320°C , holding sensible c_p reduces it by 7.09%, but reduces latent-inclusive effective specific heat by only 0.95%; enthalpy accumulated from the cold reference changes by 0.66%. Conductivity decreases by 22.48% at the same temperature. These are distinct local comparisons: the sensible-property contrast is substantially moderated when expressed as total storage through the phase interval.

This supports a storage-based interpretation of the unequal responses without making latent heat an independently isolated cause. Latent heat and phase endpoints are unchanged in every corner, and the c_p continuation begins below the solidus. It therefore also changes sensible enthalpy acquired before melting and the temperature–enthalpy inverse throughout the hotter field. The observed shortward displacement under storage holding is consistent with a changed enthalpy-transport balance, but a matched-temperature percentage alone cannot predict the global sign or magnitude. The relative sensitivities could differ with another freezing interval, phase law, source or property function; nonlinear enthalpy treatments such as Van Elsen et al. [8] likewise make phase effects conditional on the transport setting.

The retained axial-face Péclet median increases from 4.44 to 5.77 under conductivity holding, consistent with a shift toward coordinate transport relative to diffusion in the selected discrete phase region. This transport is material passage through the laser frame, not liquid convection. Moreover, the selected face population changes with the field. The diagnostic supports a shift in the model’s transport balance, rather than isolating an exclusive bulk mechanism.

The integrated flux supplies a useful counterexample to a simpler explanation. Outward diffusive power from the selected rear hot region increases from 14.724 to 14.913 W under conductivity holding. Lower conductivity therefore does not imply lower integrated outward heat flow in this comparison: temperature gradients and the temperature-selected region boundary change simultaneously. Conductivity also enters surface recovery through its primitive and inverse, so the reported surface crossings contain both the solved field response and a consistently changed recovery step. The large tail displacement survives as an observation of this model; interpreting it solely as suppressed bulk heat escape would exceed the available diagnostics.

4.4. Spatial phase structure and material time

The current-mesh B/C rear phase spans, 80.26 and 80.79 μm , coexist with passage times of 100.33 and 67.33 μs . The distinction follows from dividing spatial separation by scan speed: faster passage shortens the material interval even when the modeled spatial phase structure changes little. The associated 49.02% increase in interval-mean cooling rate is therefore a consequence of the same thresholds and coordinate transformation, rather than another independent validation endpoint. It also differs from length sensitivity: moving both rear thresholds together changes their location relative to the laser without necessarily changing their separation.

This distinction matters when comparing simulated fields with thermography. Hooper [3] measured spatial profiles and fixed-location thermal histories in Ti6Al4V, including scan events with significant transients. Lane et al. [24] provided cooling-rate exemplars using a below-solidus interval and an imaging-based rear-boundary definition. Neither is a matched measurement of the present 1350–1290 °C centerline interval. A physical comparison would require a temperature trajectory over the same interval, with scan position, exposure and observation mapping accounted for. The transformed interval is informative about this quasi-steady model but does not determine segregation, grain morphology or the alloy’s actual nonequilibrium freezing path.

The trajectory represented by a thermal history also determines its physical meaning. Hou et al. [32] followed fluid elements through a Ni–20Cr melt pool and used their position-dependent thermal excursions to model oxide evolution. Their CFD trajectories had narrower near-surface temperature peaks and hotter subsurface excursions than the unmodified conduction solution; a reduced Eagar–Tsai field was therefore scaled against CFD before being used for thermal histories. That comparison clarifies the present diagnostic: ℓ_m/v describes material translation through the assumed steady field, whereas a liquid element can move through the field along a different trajectory. Interpreting the current passage interval as fluid residence or a structure-prediction input would require an additional trajectory model and its own evidence.

4.5. *Surviving explanations and discriminating comparisons*

Three explanations remain compatible with the shape discrepancy. First, the Gaussian surface source and constant effective factor may distribute energy incorrectly across the conditions. Calibrated conduction models can transfer successfully: Ross et al. [14] inferred volumetric source parameters from solidification-boundary temperatures and interpolated their trends; Kusano and Watanabe [16] compared source families and constructed a calibrated conical-source regression for an IN738LC process window. Coleman et al. [17] calibrated a dynamic volumetric source for bare IN625 at 195 W and 800 mm/s over multiple spot sizes, then applied it to an AMB2018-01 layer case. Their phase-dependent properties and 1410 K eutectic endpoint differ from the present equilibrium interval. These successes make source choice a substantive alternative to changing conductivity alone.

Second, constitutive uncertainty and directional melt transport may affect shape and rear temperature differently. The tested scalar continuations strongly influence length while leaving the main B shape residual. This restricts those two scenarios as geometric repairs, but does not eliminate other high-temperature functions or effective directional transport. Composition-based estimates can provide physically motivated candidate functions where measurements are unavailable [33]; they cannot identify this coupon's liquid properties. A comparison using the same measured beam, phase assumptions, length target and extraction operators would clarify whether an alternative source or directional closure improves excluded width/depth together with the rear profile. Adding an independently calibrated thermal profile would distinguish parameter compensation from credible thermal agreement.

Third, observation mapping or omitted free-surface physics may contribute. The experimental radiance length and the modeled solidus-crossing proxy differ, while fusion width/depth use different populations and a passage-maximum operator. An imaging replay could assess how this mismatch affects fitted source strength before introducing additional physics. Melt-flow studies show how recoil, surface-tension-driven motion and evaporative cooling redistribute energy [5], and X-ray imaging directly reveals vapor-depression transitions [34]. Those results establish viable mechanisms under their own conditions, rather than diagnose a keyhole or flow mechanism from the present wide/shallow residual.

Hong et al. [26] illustrate a more discriminating failure test. Their Eagar–Tsai bare-track analysis uses melting-range effective properties, separately specified absorption inputs, and depth inversion to ask whether any physical absorption can reach the measured penetration; property and directional-diffusivity controls test that criterion. IN625 absorption uses an empirical relation rather than a matched in-situ measurement. The present study performs neither that inversion nor those source/transport controls. Its shape discrepancy therefore identifies a model limitation after fitting, without establishing the physical necessity of a vapor cavity. The useful next comparison is one that constrains source/measurement mapping and tests an excluded thermal or shape observable, rather than selecting a mechanism from the residual alone.

Benchmark design can make these distinctions experimentally actionable. The AM Bench 2022 IN718 track and pad measurements combine controlled power, speed and spot-size changes with thermography, cross-sectional geometry and stated uncertainty budgets [35]. Their reported pad heat buildup also shows why a single-track comparison cannot establish multi-track transfer. For the present case, a prospective test would retain one defined source and observation mapping, reserve excluded tracks or thermal profiles before fitting, and evaluate geometry together with a temperature trajectory. Such a design would test the thermal meaning of calibration rather than add another dependent quantity extracted from the same computed field.

4.6. Limits on thermal interpretation

Two constraints govern the strength of this diagnosis. The continuation functions start above supplier-table endpoints of 982 °C for k and 1093 °C for c_p , both below the solidus [27]. Neither is measured liquid behavior, and the equilibrium phase law remains assumed.

High-temperature property measurements provide a route beyond deterministic continuation scenarios. For IN718, Hosaeus et al. [36] used rapid resistive heating and time-resolved measurements to determine solid- and liquid-state thermophysical relations, including enthalpy and specific heat; conductivity near the phase transition was estimated from electrical resistivity through the Wiedemann–Franz relation. This precedent distinguishes measured storage from indirectly estimated conductivity. The alloy difference prevents substituting those values for the present IN625 inputs. Resolving which continuation is physically appropriate would require composition- and state-matched property evidence, while validating the rear thermal history would additionally require an independent temperature trajectory with the same observation definition.

The retained surface peak temperatures for LL, HL, LH and HH are approximately 2697, 3507, 2756 and 3721 °C, respectively. The particularly high held-conductivity peaks illustrate the sensitivity branches; they are outside credible physical-temperature prediction in a model omitting evaporation, melt flow and surface evolution. Numerical verification, including the simpler analytical benchmark and retained axial/domain comparisons, supports the scale of the main length and shape contrasts [31]; it supplies neither a full three-dimensional error bound for every branch nor precision for submicrometre interactions.

The quasi-steady straight-track setting also limits transfer to scan turnarounds and powder layers. Martin et al. linked scan-velocity transients to vapor-depression collapse and pore formation using X-ray imaging and multiphysics calculations [37]; those events are absent from the steady solution here. Powder introduces additional interactions: Matthews et al. related denudation to vapor flux and gas-driven entrainment [38], while recent gas–melt–particle modeling identifies entrained-particle momentum as a possible perturbation of the thermal tail [39]. These studies define mechanisms and operating conditions that must be added and assessed before extending the present bare-plate diagnosis to powder-bed defects.

The experimental comparisons are retrospective and nonblind, with one fitted length, excluded shape diagnostics and separately sourced uncertainty scales. They cannot form a joint statistical validation of the thermal field. The earned implication is consequently specific: geometric fitting remains useful, but closure response, boundary translation, phase separation and material time must be distinguished before assigning solidification meaning to the fitted solution.

5. Conclusions

The practical value of geometric calibration depends on the thermal interpretation subsequently drawn from it. For the present IN625 moving-frame enthalpy model, matching one surface length does not establish agreement of the fusion shape or remove dependence on high-temperature closure. The length-fitted condition remains 19.54 μm too wide and 4.88 μm too shallow relative to the nominal reference means; the faster fixed-parameter condition retains the same wide/shallow pattern. The observed discrepancies greatly exceed the inspected baseline axial changes, while source representation and observation mapping remain viable competing explanations.

Separating conductivity and sensible-storage continuations exposes information hidden by their joint change. Conductivity holding increases length by 43.21 μm and storage holding decreases it by 6.17 μm ; the $-2.76 \mu\text{m}$ interaction gives a net increase of 34.28 μm . Their directions persist at both tested levels of the other factor. The rear solidus supplies 98.3% of the conductivity-induced length increase, and both rear phase boundaries move much farther than their separation changes.

The principal response is therefore a displaced thermal tail. Opposing depth effects partly cancel, yet none of these continuations resolves the remaining shape discrepancy.

The spatial interval and the material interval convey different information. Rear phase spans near $80\text{ }\mu\text{m}$ at 0.8 and 1.2 m/s correspond to passage times of 100.33 and 67.33 μs , and a 49.02% difference in the derived interval-mean cooling rate. This result is the material-coordinate transformation of the modeled field. It supplies a useful thermal-history diagnostic but does not independently establish cooling-rate or microstructure accuracy.

Together, the findings support evaluating geometric calibration through separate shape, closure and material-time questions. A common rear-boundary displacement should not be read as a comparably enlarged solidification interval, and a small joint property response should not be read as insensitivity of each constituent. This distinction identifies the next evidence needed for thermal credibility: aligned source and observation definitions, credible high-temperature material data and an independent thermal observable under the stated process conditions.

Data and code availability

The LaTeX manuscript, vector figures, extracted tables and evidence inspection are available in the [manuscript package](#). The original case and numerical implementation are referenced through [SimAgent](#). Experimental sources remain the cited NIST challenge and Lane study. Publisher full-text copies used for reading are not redistributed. The package README distinguishes extracted tables and figures from large retained fields and identifies the corresponding original case locators.

Declaration of generative AI use in manuscript preparation

This first draft was prepared with OpenAI Codex assistants for evidence extraction, argument planning, English drafting and independent automated review. ResearchFlow and the Microsoft Agent Framework-based assistant provided task coordination and provenance. Scientific values were taken from located numerical and experimental sources. Figures were produced by deterministic plotting, without generative image creation. Authorship, human verification of the complete manuscript, final disclosure wording, funding and competing-interest declarations must be completed by the responsible researchers before submission.

Appendix A. Reproducible material and numerical controls

Appendix A.1. Executed material and nonlinear controls

The Anderson controls are history depth $M = 8$, initial scale $\alpha = 0.2$, regularization $w_0 = 0.01$, fixed-point residual tolerance 2×10^{-4} K in U , and an Armijo line search. The GMRES cap is 800 inner iterations with restart 80 and a 2×10^{-9} tolerance. The preconditioner is Ruge–Stuben classical multigrid with strength threshold 0.25, symmetric Gauss–Seidel smoothing, V cycles, at most 20 levels and a coarse size of 100. The executed environment uses one BLAS thread. These controls are taken from the production manifests; alternative methods present in the solver source were not used for the six retained production solutions.

Appendix A.2. Separate constant-property verification problem

The Gaussian verification uses $P = 179.2$ W, $\eta = 0.2$, $v = 0.8$ m/s and $w = 85\text{ }\mu\text{m}$, with $\rho = 8440\text{ kg m}^{-3}$, constant $c_p = 410.94444\text{ J kg}^{-1}\text{ K}^{-1}$, constant $k = 9.870588\text{ W m}^{-1}\text{ K}^{-1}$ and no latent heat. Thus $a_0 = k/(\rho c_p) = 2.845886 \times 10^{-6}\text{ m}^2/\text{s}$. Its finite-domain mesh contains 178992 cells ($113 \times 33 \times 48$), with near-source spacings of $8 \times 6 \times 2\text{ }\mu\text{m}$. Domain extents follow the production

Table A.4: Property knots loaded by the executed material routine from the general IN625 bulletin [27]. The two temperature columns describe different tables. Conductivity and sensible specific heat are interpolated independently.

T_k (°C)	k (W m ⁻¹ K ⁻¹)	T_c (°C)	c_p (J kg ⁻¹ K ⁻¹)
-157	7.2	-18	402
-129	7.5	21	410
-73	8.4	93	427
-18	9.2	204	456
21	9.8	316	481
38	10.1	427	511
93	10.8	538	536
204	12.5	649	565
316	14.1	760	590
427	15.7	871	620
538	17.5	982	645
649	19.0	1093	670
760	20.8		
871	22.8		
982	25.2		

geometry, but the adiabatic transverse and bottom boundaries truncate the semi-infinite analytical referent.

With $Q = \eta P$, the semi-infinite Neumann heat-kernel convolution for the moving Gaussian is

$$T - T_0 = \frac{Q}{\rho c_p \pi \sqrt{\pi a_0}} \int_0^\infty \frac{\exp \left[-\frac{(\xi + v\tau)^2 + y^2}{w^2/2 + 4a_0\tau} - \frac{z^2}{4a_0\tau} \right]}{\sqrt{\tau}(w^2/2 + 4a_0\tau)} d\tau. \quad (\text{A.1})$$

The archived implementation removes the surface singularity by $\tau = (w/v)s^2$ and uses adaptive quadrature. Solidus geometry is extracted at 1290 °C, with uniform-core face-average source recovery for the numerical surface. Width/depth use the monotone constant-property maximum-temperature envelope; they do not replace the general production passage observer. The original case retains the recipes `verify_constant_gaussian_frame_native61.py` and `qoi_constant_gaussian_reference_native61.py` in its collaboration directory and the corresponding `native61-constant-gaussian-frame-r1` verification reports. The quoted differences combine discretization, reconstruction and finite-domain effects, and do not bound nonlinear phase-change or physical-model error.

Appendix A.3. Scalar calibration and retained sensitivities

The calibration bracket $\eta \in [0.2, 0.4]$ has retained endpoint lengths of 234.57 and 511.68 μm . Subsequent retained trial pairs $(\eta, L/\mu\text{m})$ are (0.2898051, 359.96) and (0.2853148, 354.22), followed by the final (0.2890548519, 359.16). They establish a sampled local bracket, not global identifiability. The local secant sensitivity is approximately 1276.79 μm per unit η . Dividing B's 21.26 μm expanded uncertainty by this slope gives a scale $\Delta\eta = 0.01665$; it is not a posterior interval and is not propagated into the geometry comparisons.

For uncertainty-scale reporting, $\delta/U = (Q_{\text{model}} - \overline{Q}_{\text{source}})/U_{\text{Lane}}$. The separate U values are not specimen standard deviations. Imaging and microscopy have different operators and aggregation, so the geometry triplet is not fitted to a joint likelihood. All finite contrasts use full-precision outputs before rounding for display.

Table A.5: Retained calibrated axial sensitivities in μm , calculated as fine minus baseline. Only the near-source axial spacing changes from 4 to 2 μm ; transverse and depth spacings remain 3 and 1 μm . These contrasts do not bound every discretization component.

Condition	ΔL	ΔW	ΔD
B	+0.361	−0.009	+0.041
C	+0.199	−0.029	+0.045

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